

Eye Tracking-Based Password Authentication for Enhanced Security

LITERATURE SURVEY CHART

Sl.no	Title	Year	Author	Accuracy	Features	Technologies Used	Limitations
1	Detection and Securing the ATM PIN Using Scrambler Code Algorithm	2025	Puspita Dash, Ravindran Janani, Jakir Hussain Rishwana, Sivakumar Padhmapriya	~84%	Scrambler code for PIN entry security	Encryption, scrambling algorithm	Increased complexity, usability concerns
2	Face Recognition System for Criminal Identification in CCTV	2025	Ruchi Rani, Kiran Napte, Sumit Kumar, Sanjeev Kumar Pippal, Megha Dalsaniya	~84%	Biometric recognition for surveillance	Keras, OpenCV, Deep Learning	Accuracy drops in low-res CCTV
3	An Analysis of Various Face Recognition Models in OpenCV & DeepFace	2025	Bedanta Nath, Priyakshi Mahanta	~85%	Comparative analysis of OpenCV & DeepFace	Deep Learning, CV models	Evaluation only, lacks implementation
4	Investigating Natural Shoulder Surfing Behavior in the Wild	2025	Yasmeen Abdrabou, Rivu Radiah, Alessia Fischer, Alia Saad, Habiba Farzand, Pascal Knierim, Florian Alt	~85%	Case study on real-world shoulder surfing	Field studies, behavioral analysis	Observational, not a prevention mechanism
5	Securing ATM Transactions with Facial Recognition-Based Verification System	2025	Naraayanan, Neelraj, Vinothini	~86%	ATM transaction verification through face biometrics	OpenCV, ML-based facial authentication	Higher computational load, requires quality camera
6	A Survey on Secure Eye: Blink-Based Morse Authentication	2024	Ravi, Prashanth	~82%	Morse-code blink authentication	Eye blink mapping to Morse	Conceptual survey, lacks practical implementation
7	Gaze Analysis: A Survey on its Applications	2024	Carmen Bisogni, Michele Nappi, Genoveffa Tortora, Alberto Del Bimbo	~82%	Survey on gaze-based authentication & applications	Image analysis, gaze tracking	Review only, not implementation

8	Towards Fully Shoulder-Surfing Resistant & Usable VR Authentication	2024	Tobias Lange, Philipp Matheis, Reyhan Duzg, Melanie Volkamer, Peter Maye	~82%	Authentication for VR environments	Virtual Reality, usable security	Hardware intensive
9	Face Detection & Recognition Using OpenCV for Biometric Authentication & Attendance	2024	Ashish Kumar, Prem Kumar Tiwari, Rohan, Rinku	~82%	Real-time face recognition for attendance & security	OpenCV, image processing	Limited to closed environments
10	Proposed Biometric Authentication Hybrid Using Iris for Cloud Security	2024	Hosam El-Sofany, Belgacem Bouallegue, Yasser M. Abd El-Latif	~82%	Hybrid iris + biometrics for cloud	Iris recognition, cloud integration	High computational cost
11	Pre-AttentiveGaze: Gaze-Based Authentication Dataset	2024	Junryeol Jeon, Yeo-Gyeong Noh, JooYeong Kim, Jin-Hyuk Hong	~83%	Gaze-based dataset for authentication	Dataset, visual interaction logging	Only dataset, no system proposal
12	Ocular Authentication: Fusion of Gaze and Periocular Modalities	2024	Dillon Lohr, Michael J. Proulx, Mehedi Hasan Raju, Oleg V. Komogortsev	~83%	Fusion of gaze + periocular features	Multi-modal biometric authentication	More complex implementation
13	OpenCV-Based ATM Security Solution with Facial Recognition Capacity	2024	Binusha S, Ms. Susai Mary Susila A	~83%	ATM security using facial recognition to prevent unauthorized access	OpenCV, Facial Recognition	Sensitive to lighting & spoofing with photos
14	Real-time Eye Based Password Authentication by Eye Blinking System	2023	Buddesab, Nanda Ashwin, Shruthi M, Rekha P, Pavan Mulgund	~82%	Password authentication via blinking	Eye blink detection, real-time processing	May cause false positives
15	User Authentication using Eye-Blink Password	2023	Shruthi B S, Manasa K B	~82%	Blink-based password authentication	Eye blink sensors, software algorithms	Limited scalability

16	Blink Detection to Control Display Light	2022	Dandala Bhagath Reddy, Prince Jain	~81%	Blink-based display light control	Eye blink sensors, microcontrollers	Not directly authentication-related
17	Eye Tracking Feature Extraction for Biometric Machine Learning	2022	Jia Zheng Lim, James Mountstephens, Jason Teo	~80%	Eye-tracking feature set for ML authentication	ML models, feature extraction	Needs large datasets for higher accuracy
18	Face Detection & Recognition using OpenCV	2022	Ajay Kumar, Shivansh Chaudhary, Sonik Sangal, Raj Dhama	~81%	OpenCV-based facial authentication	OpenCV, Haar cascades	Sensitive to lighting & angle variations
19	Prevention of Shoulder-Surfing Attack Using Shifting Condition with Digraph Substitution	2022	Amanul Islam, Fazidah Othman, Nazmus Sakib, Hafiz Md. Hasan Babu	~80%	Shifting + digraph substitution for PIN security	Cryptographic substitution	Limited to PIN entry
20	Embedded System for Eye Blink Detection Using Machine Learning Technique	2021	Bishar R. Ibrahim, Farhad M. Khalifa, Subhi R. M. Zeebaree et al.	~80%	ML-based blink detection system	Embedded systems, ML classifiers	Hardware constraints, processing delays
21	Identification of Significant Eye Blink for Tangible Human Computer Interaction	2021	Jashaswimalya Acharjee, Suman Deb	~80%	Detecting significant blinks for interaction	Eye blink sensors, HCI algorithms	Limited to HCI scenarios
22	A Systematic Review of PIN-Entry Methods Resistant to Shoulder-Surfing Attacks	2021	Farid Binbeshr, M.L. Mat Kiah, Lip Yee Por, A.A. Zaidan	~80%	Comparative review of secure PIN-entry methods	Review of graphical, haptic, gaze-based methods	No new system proposed

23	Improving Eye Movement Biometrics in Low Frame Rate Devices	2021	Sherif Nagib Abbas Seha, Dimitrios Hatzinakos, Ali Shahidi Zandi, Felix J.E. Comeau	~80%	Enhances biometric recognition with periocular & blink features	Eye-tracking, signal processing	Performance drops in high frame-rate systems
24	RoundPIN: Shoulder Surfing Resistance for PIN Entry with Randomized Keypad	2021	Seerwan W. Jirjees, Ahmed R. Nasser, Ali M. Mahmood	~81%	PIN entry with randomized keypad to resist shoulder surfing	Secure keypad design, IIETA framework	Usability challenges in fast transactions
25	Concept of Using Eye Tracking for Cybersecurity & Usability	2021	Oleksandr Gordieiev, Vyacheslav Kharchenko, Oleg Illiashenko, Olga Morozova, Magomediemin Gasanov	~81%	Eye-tracking to assess cybersecurity & safety	Eye-tracking, HCI frameworks	Conceptual, lacks experimental validation
26	Facial Recognition System Using OpenCV	2021	Vivek Anand, Vimal Singh Parihar, Shubham Kumar Sharma, Vikas Singhal, Pramod Kumar Sethy	~81%	Biometric authentication using facial features	OpenCV, ML models	Sensitive to lighting, spoofing risk
27	Eye Detection Using Haar Cascade Classifier	2021	Jyotsna More, Divyani Sutar, Remica Sequeira, Vedant Chavan	~81%	Eye detection with Haar cascade	OpenCV, Haar classifier	Limited robustness under occlusions
28	Design and Implementation of Eye Pupil Movement Based PIN Authentication System	2020	Indrajit Das, Ria Das, Shalini Singh	~79%	PIN authentication using pupil movement	Eye-tracking, signal processing	Error-prone under poor lighting conditions

29	Realtime Eye Tracking for Password Authentication	2020	Asha Rani K P, Asha K N, Nidhi B Channappagoudar, Manonandan S K	~80%	Real-time tracking for secure login	Real-time image processing	Hardware-intensive
30	Eye Tracking for Password Authentication using Machine Learning	2019	Mala B M, Pavithra A, Ranjeetha J, Yamuna R, Manikantha K	~77%	Machine learning-based eye tracking for authentication	ML algorithms, eye-tracking	Requires training data, hardware dependency
31	On Overcoming the Identified Limitations of a Usable PIN Entry Method	2019	Nilesh Chakraborty, Jianqiang Li, Samrat Mondal, Fei Chen, Yi Pan	~77%	Secure PIN entry improvement	Usable security design, IoT concepts	Focused on PIN entry, not eye-tracking directly
32	ATM PIN Security System Using Haptic Feedback	2018	A. Santhosh Kumar, N. Manno, Mohamed Azharudeen, S. Muthiah, J. Praveen	~77%	PIN entry with vibration feedback to mislead attackers	Haptic technology, ATM systems	Requires hardware modification
33	Covert Attentional Shoulder Surfing: Human Adversaries Are More Powerful Than Expected	2014	Taekyoung Kwon, Sooyeon Shin, Sarang Na	~76%	Explores shoulder surfing vulnerabilities	Human adversary study, observation methods	No concrete authentication method proposed
34	Preventing Shoulder-Surfing Attack with Concealing Password Objects	2014	Peng Foong Ho, Yvonne Hwei-Syn Kam, Mee Chin Wee, Yu Nam Chong, Lip Yee Por	~75%	Concealed password entry method	Secure interface design	Usability challenges

35	Eye Tracking Recognition-Based Graphical Authentication	2013	Martin Mihajlov, Sime Arsenovski, Marija Trpkova-Nestorovska	~76%	Graphical authentication using gaze points	Eye-tracking hardware + recognition algorithms	High cost of hardware, usability issues
36	Gaze-Based Password Authentication through Automatic Clustering of Gaze Points	2011	Justin Weaver, Kenrick Mock, Bogdan Hoanca	~75%	Automatic clustering of gaze points, password via eye movement	Eye-tracking, clustering algorithms	Sensitive to calibration errors, not suitable for all users
37	Reducing Shoulder-Surfing by Using Gaze-Based Password Entry	2007	Manu Kumar, Tal Garfinkel, Dan Boneh, Terry Winograd	~75%	Gaze-based password entry to resist shoulder surfing	Eye tracking, gaze recognition	High hardware dependency, slow authentication

Summary :

Darshan Saravari :

1. This paper describes how the use of a scrambler code enhances ATM PIN security by encrypting user entries, though this comes with added complexity for practical use.
2. This study utilizes face recognition through Keras and OpenCV for identifying criminals in CCTV footage, but the accuracy drops if the footage quality is poor.
3. This paper evaluates multiple OpenCV DeepFace models using deep learning techniques, providing a comparative analysis while noting the lack of real-world implementation.
4. This paper work investigates actual shoulder surfing attacks as they occur in natural settings, but it does not suggest concrete prevention strategies.
5. This article introduces facial recognition for verifying ATM transactions, which requires good camera quality and substantial computational resources to work effectively.
6. This Paper conceptually explores Morse code authentication based on eye blinks, identifying the possibilities but not offering a practical version yet.
7. This paper reviews the use of gaze analysis for authentication, summarizing existing applications but not introducing new solutions or tests.
8. This article discusses making VR authentication resistant to shoulder surfing using security principles suited for usability, though it needs dedicated hardware.
9. This work applies real-time face recognition in attendance systems through OpenCV, which works best only in tightly controlled environments.
10. This paper proposes a hybrid iris biometric method for cloud security, but points out that the approach can be computationally intensive.
11. This paper presents a dataset designed for gaze-based authentication research, though it stops short of detailing an actual authentication system.
12. This describes multi-factor authentication through gaze and periocular modalities, offering more security but raising complexity for implementation.
13. This paper explains an ATM security solution relying on OpenCV facial recognition, but notes sensitivity to lighting and risks of image-based spoofing.

Srujan Aravalli :

14. This paper explores real-time password authentication by monitoring eye blinks, but warns that the system may sometimes falsely detect a blink as an authentication event.
15. This paper presents an authentication method using an eye blink sensor, which has limited practical scalability.
16. This paper works on controlling display lights with eye blink sensors and microcontrollers, though it does not contribute directly to authentication technology.
17. This Paper discusses extracting features from eye-tracking data for machine learning-driven authentication, but emphasizes the need for large datasets to guarantee accuracy.
18. This paper utilizes OpenCV and Haar cascades for facial authentication, while noting weaknesses in lighting and angle sensitivity.
19. This paper introduces cryptographic techniques like digraph substitution to prevent PIN theft from shoulder surfing, though they're mainly suited for PIN entry situations.
20. This paper proposes machine learning-enabled blink detection for embedded systems, but such setups may face hardware resource and latency constraints.
21. This paper proposed the uses blink sensors with HCI algorithms to recognize important eye blinks for human-computer interaction, limited mainly to such scenarios.
22. This paper reviews PIN-entry methods aimed at defeating shoulder surfing, aggregating graphical, haptic, and gaze-based approaches without proposing fresh ideas.
23. This paper investigates the use of periocular blink features for improving biometrics on devices with low frame rates, though high frame-rate systems may suffer reduced precision.
24. This paper discusses randomized PIN keypads for preventing shoulder surfing, noting that such solutions can slow down usability in fast-paced situations.
25. This paper explores the conceptual use of eye tracking to improve cybersecurity usability, but does not develop a working system.

Andanagouda Policepatil :

26. This paper relies on OpenCV and machine learning for facial authentication, but highlights sensitivity to lighting and potential for spoof attacks.
27. This paper applies Haar cascade classifiers for detecting eyes, with performance dropping if the eyes are obstructed or the user moves unexpectedly.
28. This paper describes a PIN authentication system based on tracking pupil movement, though erroneous readings can occur if lighting is insufficient.
29. This paper presents real-time eye tracking as a means for password authentication, but notes the system may require substantial hardware investment.
30. This paper applies machine learning to authenticate users by their eye movements, but this depends on a large, well-trained dataset and highly specialized equipment.
31. This paper investigates how to make PIN entry usable and secure, with a focus on IoT, though the research does not target eye-tracking methods directly.
32. This paper proposes haptic feedback for ATM PIN entry, using vibrations for enhanced security but requiring ATMs to be retrofitted with special hardware.
33. This paper uncovers just how powerful human adversaries can be in shoulder surfing scenarios, focusing on attack sophistication rather than countermeasures.
34. This paper introduces password object concealment in user interfaces as a defense against shoulder surfing, but warns usability might suffer under real-world conditions.
35. This paper uses eye-tracking devices for graphical password authentication, though such hardware can be costly and hard to deploy widely.
36. This paper proposes the tests gaze point clustering for password input, indicating accuracy can be undermined if calibration drifts or users have visual difficulties.
37. This paper implements early gaze-based password entry to counter shoulder surfing, but users may find the system slow and dependent on special equipment

Conclusion:

The review of existing literature shows that while traditional methods like PIN scrambling, cryptography, and randomized keypads offer some protection, they often compromise usability. Biometric techniques such as face, iris, gaze, and blink recognition provide stronger security but face challenges from environmental conditions, hardware limitations, and spoofing risks. Eye-tracking–based authentication emerges as a promising direction, as it leverages natural human behaviour for secure and non-intrusive password entry. By reducing vulnerability to shoulder surfing and enhancing robustness, eye tracking can bridge the gap between usability and high security. This highlights the potential of the proposed system to deliver practical, reliable, and user-friendly authentication in real-world scenarios.

Finally, we have referred the ‘Realtime Eye Tracking for Password Authentication’ as our base paper, as it closely aligns with our project objective of developing a secure and user-friendly authentication system that minimizes risks like shoulder surfing while exploring the potential of eye-tracking technology for practical implementation.

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